Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 2:15 PM–3:30 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 READ & REASON

Rearrange A = L × W to find W.

A. W = A × L

B. W = L ÷ A

C. W = A − L

D. W = A ÷ L

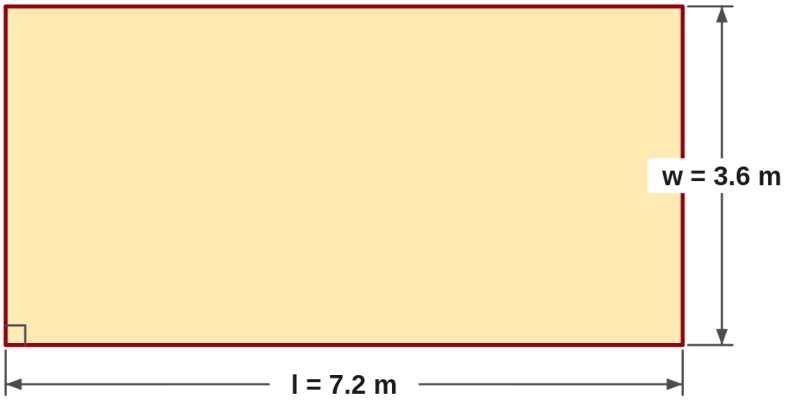
Choice (A–D) _____ **Working / reason** _____

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

02 A new floor keeps the 3.6 m width shown in Shape Canvas but has area 43.2 m². Find its new length.



Shape Canvas

Answer (m)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

03 Which Numeracy Hub calculator entry correctly finds W from P = 2(L + W), P = 30 and L = 9?

A. 30 × 2 − 9

B. 30 ÷ 2 − 9

C. 30 − 2 × 9

D. 30 ÷ (2 − 9)



Numeracy Hub

Choice (A–D)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

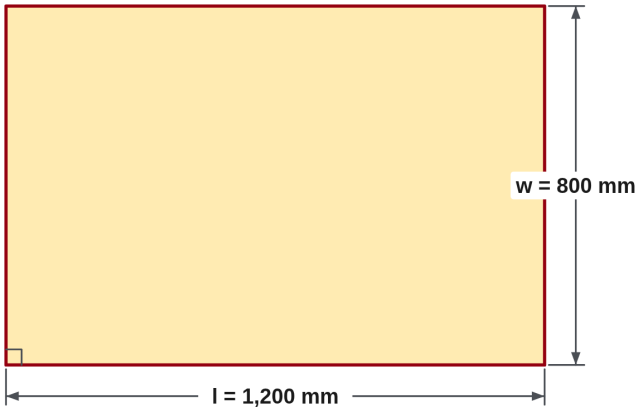
04 A rectangle’s area is 960000 mm². Its width is 800 mm. Which rearrangement gives the length shown in Shape Canvas?

A. L = 960000 ÷ 800

B. L = 960000 × 800

C. L = 800 ÷ 960000

D. L = 960000 − 800



Shape Canvas

Choice (A–D)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

05 How should a calculated missing dimension be checked?

A. Repeat the same wrong entry twice

B. Compare only the last digit

C. Delete the known dimensions

D. Substitute it into the original formula

Choice (A–D)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

06 CALCULATE

A rectangle has area 18 m² and width 2.4 m. Find its length.

Answer (m) | **Working / reason**

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

07 P = 2(L + W), P = 38 m and L = 12 m. Find W.

Answer (m) | **Working / reason**

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

08 A = b × h / 2, A = 24 m² and b = 8 m. Find h.

Answer (m) | **Working / reason**

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

09 V = L × W × H, V = 4.8 m³, L = 4 m and W = 2 m. Find H.

Answer (m) | **Working / reason**

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

10 T = 5l + 25, with T = 1825 mm. Find l in millimetres.

Answer (mm)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

11 APPLY & CHECK

A student finds W = P − 2L from P = 2(L + W). Why is that incomplete?

A. P must be squared

B. L should be added twice

C. The result is already W

D. P − 2L equals 2W, so divide by 2

Choice (A–D)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

12 A 6 m by W rectangle has area 33 m². Find W, then use multiplication to check.

Answer (m)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

Rearranging formulas and checking backwards

Day 15 • 2026-11-25 • 15 questions

13 A missing width calculated from positive area and positive length is negative. What should happen next?

A. Round it to zero

B. Recheck the rearrangement and arithmetic

C. Use its negative value on the drawing

D. Ignore the sign without checking

Choice (A–D)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

14 A rectangular frame has P = 17.6 m and W = 3.4 m. Find L.

Answer (m)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204

15 For V = LWH, which expression isolates H correctly?

A. H = V ÷ L × W

B. H = L × W ÷ V

C. H = V − L − W

D. H = V ÷ (L × W)

Choice (A–D)

Working / reason

Textbook sections 8.3–8.5 pp.195–200; sections 8.7 pp.203–204